

Core Patterns Checklist (Part 2)

4. Linked Lists

Focus: Pointer manipulation, fast & slow pointers (Floyd's cycle detection), and in-place reversal/merging without extra memory.

- [206. Reverse Linked List](#) (Easy) — *The foundational iterative pointer reversal.*
- [876. Middle of the Linked List](#) (Easy) — *Fast and slow pointer standard traversal.*
- [141. Linked List Cycle](#) (Easy) — *Floyd's Cycle-Finding Algorithm (Tortoise and Hare).*
- [21. Merge Two Sorted Lists](#) (Easy) — *Dummy node usage for clean list merging.*
- [160. Intersection of Two Linked Lists](#) (Easy) — *Two-pointer length synchronization.*
- [234. Palindrome Linked List](#) (Easy) — *Combining middle-finding and reversal techniques.*
- [203. Remove Linked List Elements](#) (Easy) — *Handling head deletions and standard traversal modifications.*
- [19. Remove Nth Node From End of List](#) (Medium) — *One-pass delayed two-pointer approach.*
- [142. Linked List Cycle II](#) (Medium) — *Math-based cycle entry point detection.*
- [2. Add Two Numbers](#) (Medium) — *Simulating basic addition with carryover across two lists.*
- [143. Reorder List](#) (Medium) — *Splitting, reversing, and merging halves (combines 3 basic patterns).*
- [138. Copy List with Random Pointer](#) (Medium) — *Deep copy using interleaved nodes or a HashMap.*

5. Stacks & Monotonic Stacks

Focus: LIFO operations, string parsing, expression evaluation, and maintaining monotonic (strictly increasing/decreasing) states.

- [20. Valid Parentheses](#) (Easy) — *The classic LIFO bracket matching problem.*
- [496. Next Greater Element I](#) (Easy) — *Introduction to the monotonic decreasing stack pattern.*
- [1047. Remove All Adjacent Duplicates In String](#) (Easy) — *String manipulation using a stack for immediate state checks.*
- [682. Baseball Game](#) (Easy) — *Basic simulation of undo/record operations.*
- [232. Implement Queue using Stacks](#) (Easy) — *Amortized $O(1)$ operations using two stacks.*
- [155. Min Stack](#) (Medium) — *Parallel state tracking for $O(1)$ minimum retrieval.*
- [150. Evaluate Reverse Polish Notation](#) (Medium) — *Evaluating postfix expressions.*
- [71. Simplify Path](#) (Medium) — *Directory traversal and string tokenization via splitting.*
- [394. Decode String](#) (Medium) — *Nested state management using a stack.*
- [739. Daily Temperatures](#) (Medium) — *The definitive monotonic stack problem for finding next greater values.*
- [402. Remove K Digits](#) (Medium) — *Greedy monotonic stack for forming the smallest number.*
- [84. Largest Rectangle in Histogram](#) (Hard) — *Advanced monotonic stack to calculate boundaries.*